Investment Portfolio & Risk Analysis

Comparing Three Asset Allocation Strategies, 2016–2025

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1. Executive Summary

This project compares three hypothetical portfolios combining U.S. equities (IVV), international equities (IXUS), and U.S. Treasury bonds (IEF). Each portfolio starts with $10,000 and rebalances monthly to fixed target weights. The analysis covers January 2016 through December 2025 and evaluates compounded returns, annualized volatility, month-end drawdowns, and Sharpe ratios.

The Growth portfolio achieved the highest ending value of $28,216 and a compound annual growth rate （CAGR） of 10.93%. It also experienced the highest annualized volatility of 11.93% and the largest month-end maximum drawdown of 22.86%. The Conservative portfolio had lower volatility of 7.57%, but its CAGR was also lower at 6.19%. The Balanced portfolio fell between these two strategies.

Within this historical sample, increasing equity exposure was associated with higher returns and higher risk. The Growth portfolio also recorded the highest annualized Sharpe ratio at 0.76. These results describe the selected historical period and do not establish which allocation will perform best in the future.

2. Portfolio Design

The three portfolios use the following target allocations:

Asset	Conservative	Balanced	Growth
U.S. equities — IVV	30%	45%	60%
International equities — IXUS	10%	15%	20%
U.S. Treasury bonds — IEF	60%	40%	20%
Total	**100%**	**100%**	**100%**

Each portfolio is rebalanced monthly. The simulation assumes no additional contributions or withdrawals and excludes transaction costs and taxes. The Balanced portfolio serves as an internal reference for comparing outcomes.

3. Data and Methodology

The analysis uses monthly adjusted closing prices for IVV, IXUS, and IEF from Yahoo Finance. Prices are taken from the last trading day of each month, from December 2015 through December 2025, producing 120 monthly returns for January 2016 through December 2025. Adjusted prices account for distributions and stock splits.

Monthly asset returns are calculated as the current month's adjusted closing price divided by the previous month's adjusted closing price, minus one. Portfolio returns are calculated as the weighted sum of asset returns using the target allocations. Applying the same target weights each month represents monthly rebalancing. Portfolio values are compounded from an initial investment of $10,000.

Performance is evaluated using the following measures:

- **Compound annual growth rate （CAGR）:** The annualized compounded return over the ten-year period.
- **Annualized volatility:** The sample standard deviation of monthly portfolio returns multiplied by √12.
- **Maximum drawdown:** The largest percentage decline from a previous portfolio peak, measured using month-end values.
- **Annualized Sharpe ratio:** The mean monthly excess return divided by the sample standard deviation of monthly excess returns, multiplied by √12. Monthly excess returns are calculated by subtracting the corresponding monthly risk-free return from portfolio returns.

Monthly risk-free returns are sourced from the RF series in the Kenneth French Data Library and matched to portfolio returns by calendar month. Asset correlations are calculated using monthly returns over the same period. All calculations are performed in Excel using unrounded values; displayed results are rounded for presentation.

4. Results and Analysis

The three portfolios show a clear historical trade-off between return and risk. Higher equity allocations produced greater ending wealth, alongside higher volatility and larger maximum drawdowns.

Metric	Conservative	Balanced	Growth
Ending Value	$18,225.14	$22,782.67	$28,216.20
Total Return	82.25%	127.83%	182.16%
CAGR	6.19%	8.58%	10.93%
Annualized Volatility	7.57%	9.53%	11.93%
Maximum Drawdown	−19.21%	−21.03%	−22.86%
Annualized Sharpe Ratio	0.56	0.70	0.76

The Growth portfolio generated $5,433.54 more ending wealth than the Balanced portfolio. Its CAGR was 2.35 percentage points higher, while annualized volatility was 2.39 percentage points higher. Its Sharpe ratio of 0.76 also exceeded the Balanced portfolio's 0.70, indicating higher historical excess return per unit of excess-return volatility.

The Conservative portfolio reduced annualized volatility to 7.57%, but still experienced a maximum month-end drawdown of 19.21%. Its higher Treasury allocation therefore reduced risk within this sample without eliminating substantial losses.

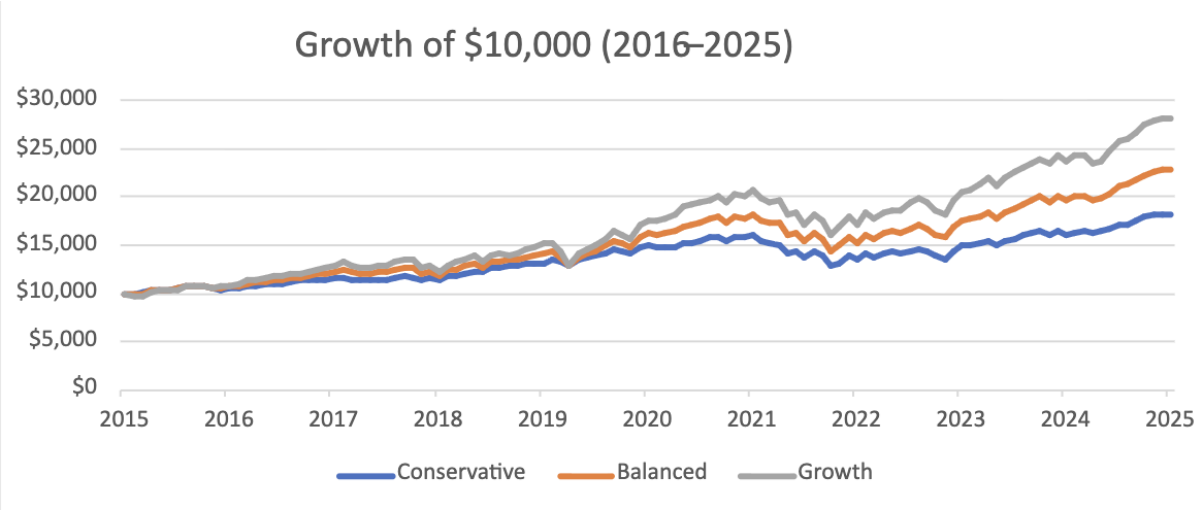


Figure 1. Growth of an initial $10,000 investment with monthly rebalancing, January 2016–December 2025. The starting value is recorded on December 31, 2015.

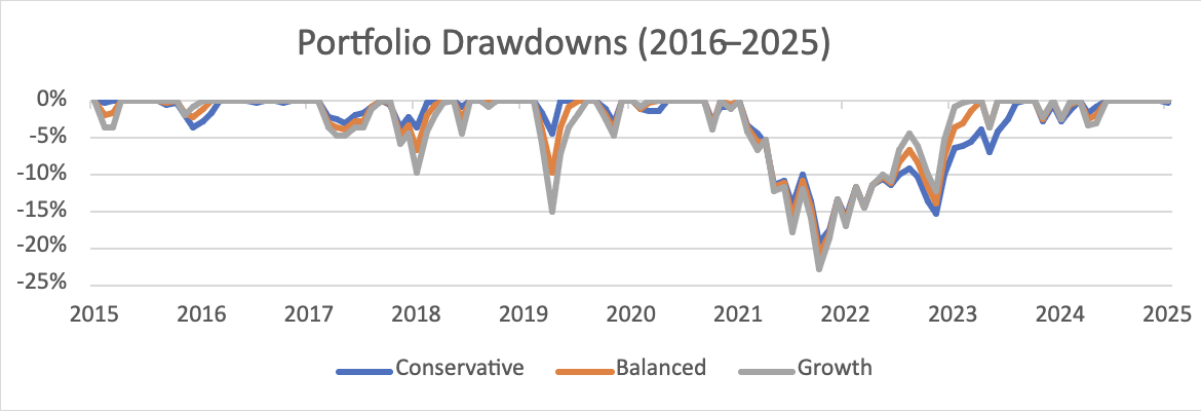


Figure 2. Portfolio drawdowns relative to previous month-end peaks. Negative values indicate declines from those peaks.

Monthly return correlations provide additional context for diversification. IVV and IXUS had a correlation of 0.84, indicating substantial co-movement between U.S. and international equities. IEF had lower correlations with IVV and IXUS, at 0.15 and 0.22, respectively. These lower correlations supported diversification over the full sample but did not guarantee protection during every market decline.

5. Conclusion

Across the three allocations tested, the Growth portfolio delivered the highest historical return and Sharpe ratio, but also the highest volatility and largest maximum drawdown. The Conservative portfolio reduced volatility at the cost of lower compounded growth, while the Balanced portfolio produced intermediate results.

The analysis demonstrates how asset allocation affects both wealth accumulation and downside exposure. It does not identify a universally optimal portfolio: the three strategies were predefined rather than optimized, and their relative performance depends on the selected sample period.

6. Limitations and Further Analysis

This analysis covers one ten-year historical period and three ETFs. Results may differ across other periods, asset classes, and market conditions. Full-period correlations can also conceal changes in diversification benefits during periods of market stress.

The simulation excludes transaction costs and taxes, assumes monthly rebalancing, and reports nominal returns without adjusting for inflation. Maximum drawdowns are measured using month-end values and may therefore understate losses experienced within a month. Sharpe ratios use square-root-of-time annualization, which assumes that serial correlation in monthly excess returns is

negligible.

Further analysis could compare alternative rebalancing frequencies, examine rolling returns and correlations, and incorporate trading costs. Testing additional historical periods would help assess whether the observed differences persist beyond this sample.

7. Data Sources

- Yahoo Finance. Historical adjusted closing prices for IVV, IXUS, and IEF, December 2015–December 2025.
- Kenneth French Data Library. Fama/French Three Factors monthly dataset, RF series, January 2016–December 2025.
- Author’s calculations in Excel. Supporting workbook: Portfolio_Risk_Analysis.xlsx.